

Improvement of Parkinson's Disease Motor Symptoms and Prevention of Dopaminergic Neuronal Degeneration by CJRB-302, a Live Biotherapeutic Product (LBP), derived from Healthy Human Gut Microbiota



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Introduction. Parkinson's disease (PD) is a progressive neurodegenerative disorder characterized by motor symptoms such as tremor, slow movements, and postural instability, along with various non-motor symptoms^{1,2}. Pathological hallmarks include Lewy bodies and loss of dopaminergic neurons in the substantia nigra pars compacta (SNpc)^{1,2}. In recent years, accumulating evidence suggests that the microbiota-gut-brain axis, a complex bidirectional pathway, plays a role in PD pathogenesis. Altered gut microbiota in patients with PD have been observed, including an increase in *Bifidobacterium*, *Lactobacillus*, and *Akkermansia*, and a decrease in *Prevotella*, *Roseburia*, *Blautia*, and *Faecalibacterium*³. In preclinical studies, fecal microbiota transplantation (FMT) from patients with PD to alpha-synuclein (α Syn) overexpressing (ASO) mice exacerbates motor deficits and α Syn pathology than wild-type mice⁴. Clinical FMT studies have shown a 5 to 13-point improvement in UPDRS motor scores, highlighting the critical role of the microbiota-gut-brain axis in PD⁵⁻⁷. However, FMT presents various disadvantages like the pathogen transfer potential and low reproducibility. Given that, there is a definite need to develop controlled and reproducible microbiome-based therapies for PD. In this study, we investigated the therapeutic potential of CJRB-302, a live biotherapeutic product (LBP) identified from healthy human gut microbiota.

Methods. **Screening & in vitro studies.** 86 strains were selected from 1,943 healthy gut-derived strains through literature and *in silico* screening. These were evaluated for antioxidant, anti-inflammatory, and neuroprotective effects *in vitro*. **In vivo studies.** Mice injected with 1-Methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP) were treated with CJRB-302. Motor deficits and tyrosine hydroxylase (TH)⁺ cell counts were assessed. Fecal samples were collected and analyzed at the end of the study. Mice with unilateral AAV- α Syn-A53T injection were evaluated for motor deficits, TH⁺ and pS129⁺ cell counts. Liver toxicity was also assessed.

Findings. CJRB-302 reduced α Syn-induced inflammatory responses and MPP⁺-induced cytotoxicity in differentiated SH-SY5Y cells, while increasing dopaminergic markers including TH and VMAT2. In mice, orally administered CJRB-302 significantly alleviated dopaminergic cell loss in the SNpc and motor deficits caused by MPTP injection. Accordingly, CJRB-302 shifted the gut microbiota towards a profile observed in healthy individuals, correlating with improved neuronal pathology, motor function, and intestinal Nurr1 protein levels. Notably, in AAV- α Syn-induced mice, oral CJRB-302 reduced α Syn pathology, motor deficit, and dopaminergic cell loss in the SNpc without exhibiting toxicity. Ongoing research aims to explore the mechanisms behind CJRB-302's beneficial effects to elucidate its mode of action in PD pathogenesis. Moreover, clinical trials are imperative to establish the efficacy and safety of CJRB-302 in human PD patients.

Conclusion. This study supports CJRB-302 as a promising new drug candidate for the development of innovative PD therapies.

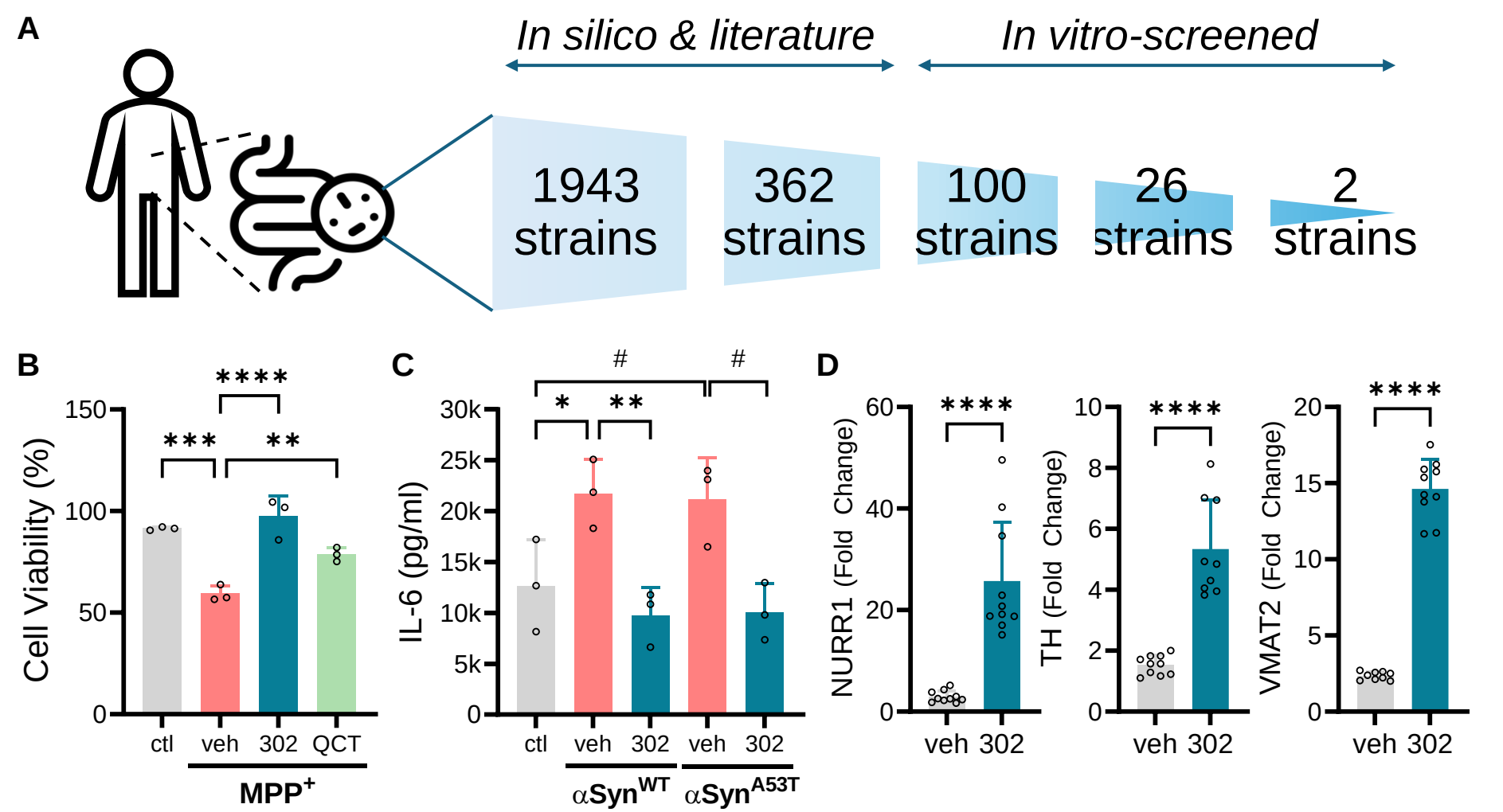


Figure 1. Selection of CJRB-302 as Parkinson's disease pipeline
(A) Screening procedures in CJRB-302 selection. (B) Neuroprotective effects of CJRB-302 supernatants in MPP⁺-induced cytotoxicity in differentiated SH-SY5Y. (C) Effects of CJRB-302 supernatants in α Syn-induced IL-6 secretion in U-373 and SH-SY5Y co-culture (D) Upregulated expression of Nurr1 and target gene transcripts in SH-SY5Y. Error bars indicate the mean $\pm$ S.D. *p < 0.05, **p < 0.01, ***p < 0.001, and ****p < 0.001 determined by one-way ANOVA with Dunnett's post hoc analysis for (B) and (C), unpaired student's t-test for (D).

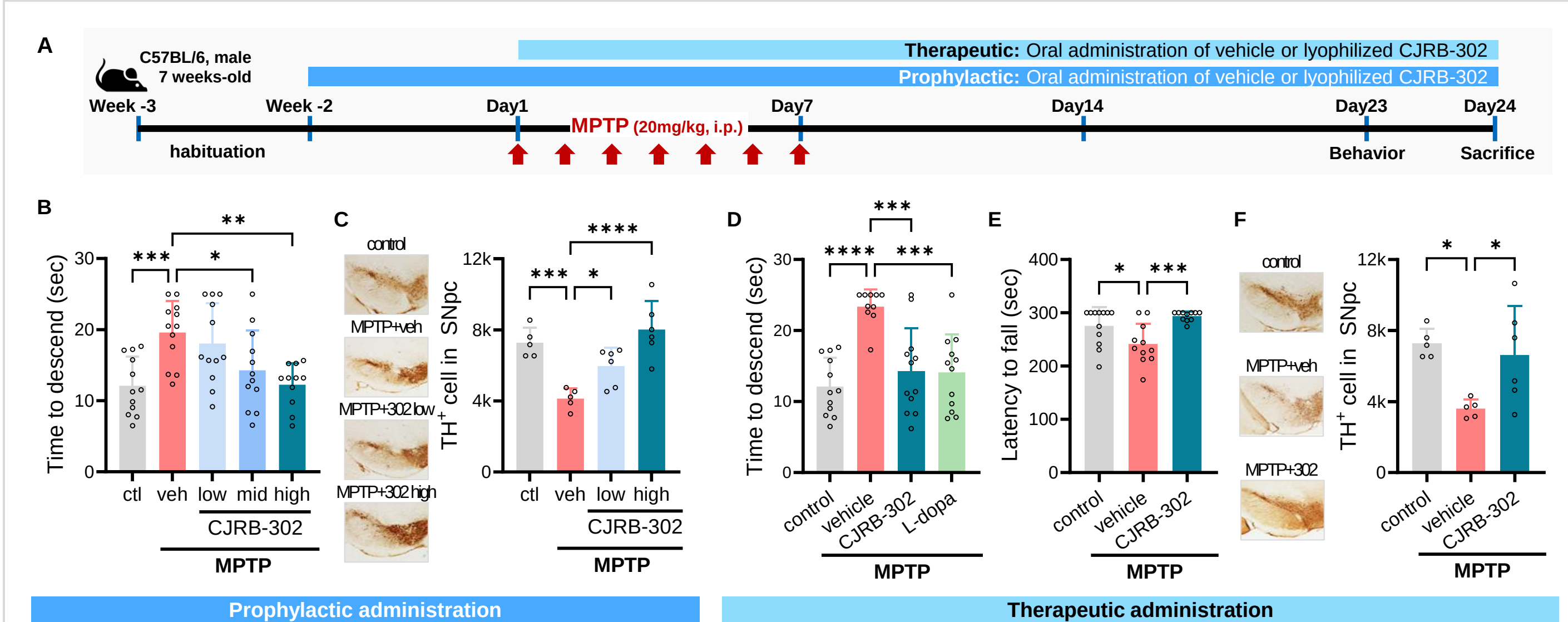


Figure 2. Oral administration of CJRB-302 ameliorates motor deficits and neuronal loss in MPTP-induced PD mouse model.
(A) Schematic illustration of CJRB-302 administration in MPTP-induced PD mouse model. (B-C) CJRB-302 administered group displayed enhanced performance in pole test (B) and higher nigral TH⁺ cell number (C) in prophylactic administration schedule. (D-F) CJRB-302 administered group displayed enhanced performance in pole test (D), rotar test (E), and higher nigral TH⁺ cell number (F) in therapeutic administration schedule. Error bars indicate the mean $\pm$ S.D. *p < 0.05, **p < 0.01, ***p < 0.001, and ****p < 0.001 determined by one-way ANOVA with Dunnett's post hoc analysis.

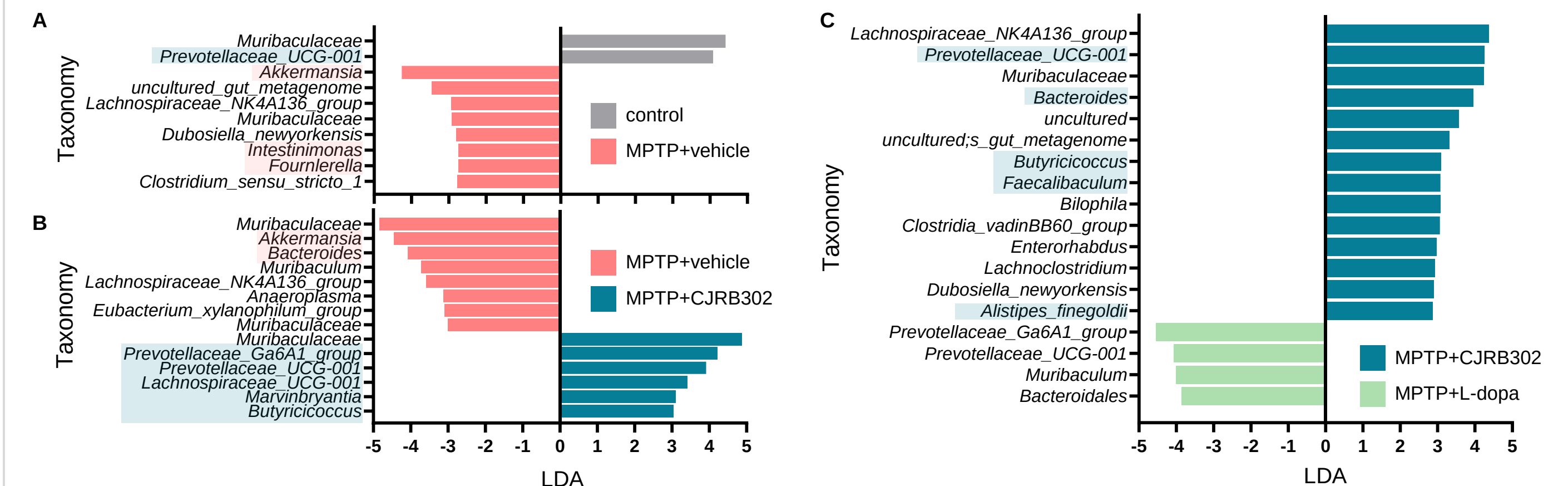


Figure 3. Oral administration of CJRB-302 to MPTP-injected mice altered PD-patient-like microbiota composition by increasing bacteria levels that are decreased in patients with PD. Linear Discriminant Analysis Effect Size (LEfSe) identifies differences in abundant taxa in the feces. The bar plot represents the log-fold differences between treatment groups, based on effect size. B_ refers to Bacteroidota, D_ to Desulfofacterota, F_ to Firmicutes, P_ to Proteobacteria, and V_ to Verrucomicrobiota, each representing a specific bacterial phylum.

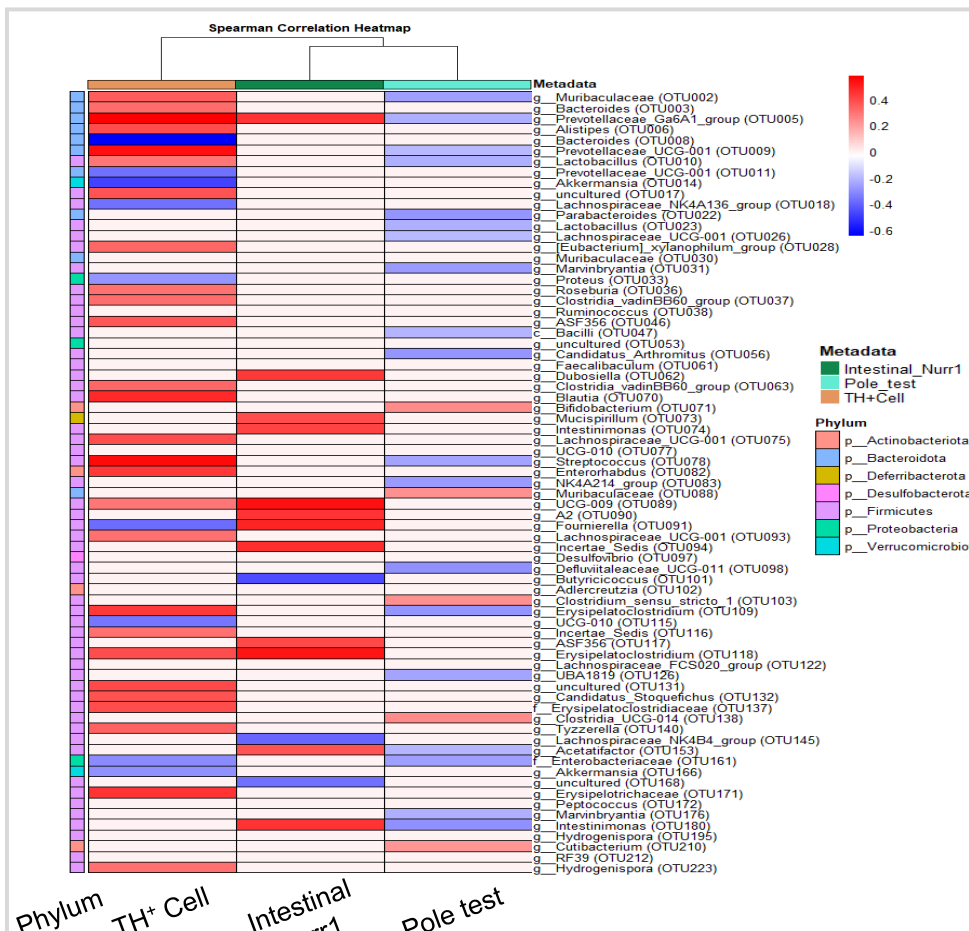


Figure 4. Spearman correlations reveal that the efficacy of CJRB-302 positively or negatively correlates with the abundances of bacteria decreased or increased in PD, respectively. One to four asterisks denote significant correlation at 0.05, 0.01, 0.001, and 0.0001 levels, respectively.

Table 1. Summary of Spearman correlations			
Metadata	Taxonomy	spearmanCoef	Significance
TH+cell	<i>g__Bacteroides</i> (OTU008)	-0.649327	****
	<i>g__Prevotellaceae_Ga6A1_group</i> (OTU005)	0.576174	****
	<i>g__Streptococcus</i> (OTU078)	0.54776	****
	<i>g__Muribaculaceae_UCG-001</i> (OTU009)	0.53084	***
	<i>g__Akkermansia</i> (OTU014)	-0.482841	***
Intestinal Nurr1	<i>g__Blautia</i> (OTU070)	0.478081	***
	<i>f__Butyricicoccaceae; g__UCG-009</i> (OTU089)	0.533864	**
	<i>g__Erysipelatoclostridium</i> (OTU118)	0.527004	**
	<i>g__Fournierella</i> (OTU091)	0.481798	**
Pole_test	<i>g__Parabacteroides</i> (OTU022)	-0.28312	**
	<i>g__Muribaculaceae_UCG002</i>	-0.264483	**
	<i>g__Lactobacillus</i> (OTU010)	-0.23069	*
	<i>g__Prevotellaceae_Ga6A1_group</i> (OTU005)	-0.229253	*

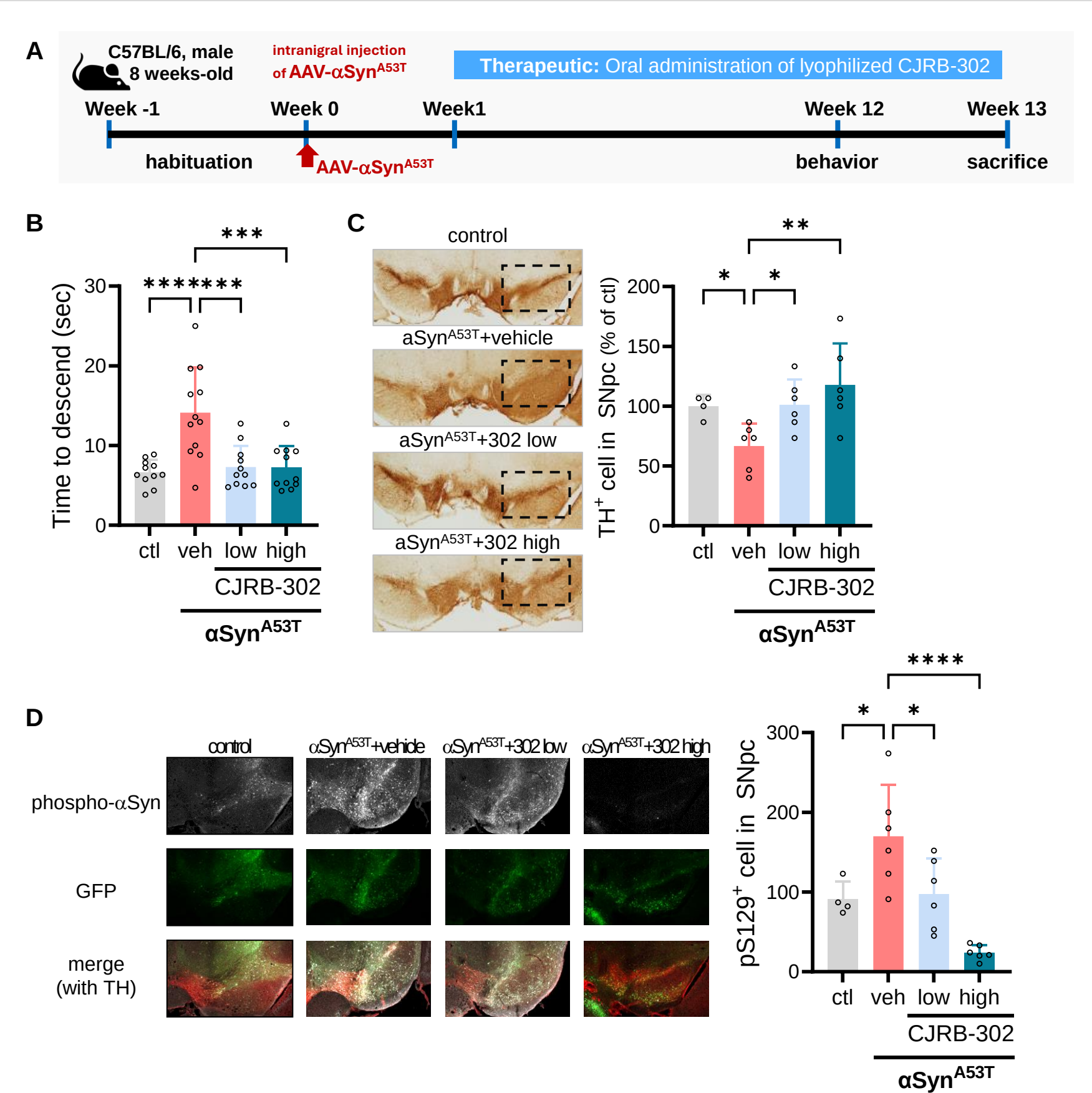


Figure 5. CJRB-302 rescues PD pathology in α Syn-induced PD mouse model.
(A) Schematic illustration of CJRB-302 administration in α Syn-induced PD mouse model. (B-D) CJRB-302 administered group displayed enhanced performance in pole test (B), higher nigral TH⁺ cell number (C), significantly diminished pS129 α Syn⁺ cells in the SNpc (D) of α Syn^{A53T} mice. Error bars indicate the mean $\pm$ S.D. *p < 0.05, **p < 0.01, ***p < 0.001, and ****p < 0.001 determined by one-way ANOVA with Dunnett's post hoc analysis for (B) and (D), and Holm-Sidak's multiple comparison test for (C).

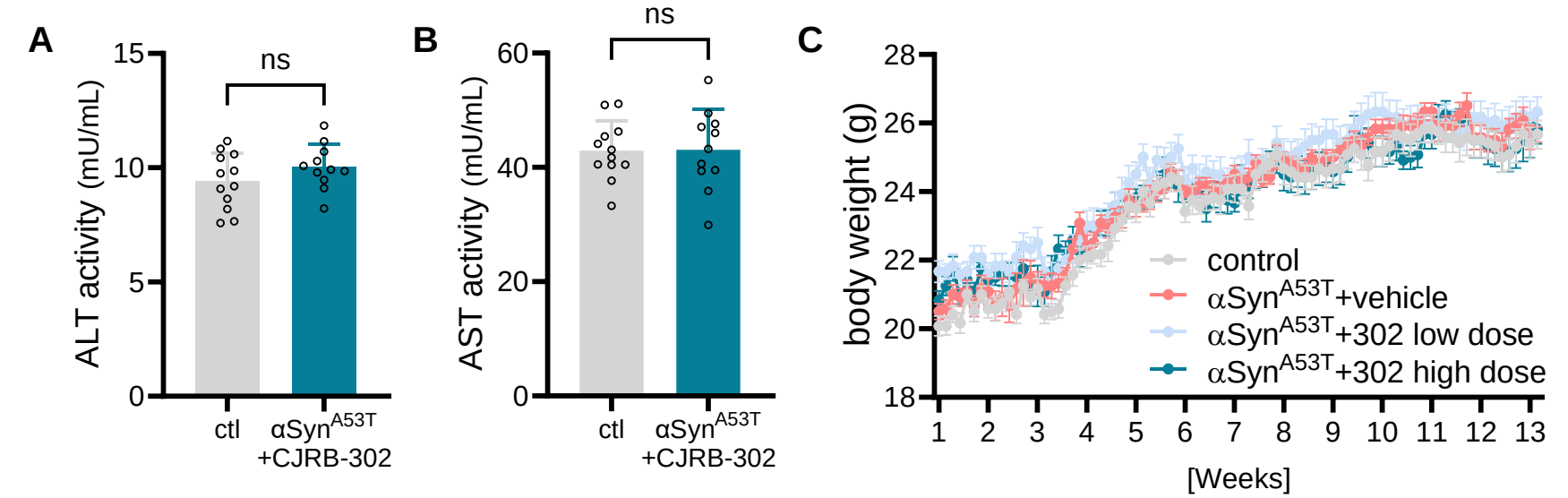


Figure 6. CJRB-302 has no adverse effects on general health.
(A-B) Non-significant difference between ALT activity (A) AST activity in the serum of control mice and CJRB-302 administered α Syn^{A53T} mice. (C) Changes in body weight recorded from the start of dosing at Week1 until the termination of the study at Week 13 in α Syn-induced PD mouse model. Error bars indicate the mean $\pm$ S.D. *p < 0.05, **p < 0.01, ***p < 0.001, and ****p < 0.001 determined by unpaired student's t-test.

References

1. Fröhlich, F. Chapter 23 – Parkinson's Disease. 2016; pp.291-296. *Academic Press*. ISBN 9780128015605; 2.Hijaz and Volpicelli-Daley, *Mol Neurodegener.* 2020;15(1):19; 3. Lorente-Picon and Laguna, *Biomolecules.* 2021;11(3):433; 4. Sampson et al., *Cell.* 2016;167(6):1449-1480; 5. Segal et al., *Clin Neurol Neurosurg.* 2021;207:106791; 6. Cheng et al., *Gut. Microbes.* 2023;15(2):2284247; 7. Bruggeman et al., *EClinicalMedicine.* 2024; 71:102563